8.3 Proving Triangle Similarity by SSS and SAS

Name _____ Period ____ Date _____

Geometry 8.3 · Student sheet

I can…

- compare the ratios of corresponding sides to decide whether two triangles are similar (SSS).
- use an included angle and two proportional sides to show two triangles are similar (SAS).
- find the value of a variable that makes two triangles similar.

Vocabulary · fill in the missing word

SSS Similarity Theorem — if the _____ side lengths of two triangles are proportional, the triangles are similar

SAS Similarity Theorem — if an angle of one triangle is congruent to an angle of another and the sides including the angles are _____, the triangles are similar

Included angle — the angle _____ by two sides

Part 1 · Discovery: build it from sides

SSS. Partner A constructs a triangle with sides 4, 5, and 7 cm. Partner B constructs one with sides 8, 10, and 14 cm. Measure the angles.

Triangle	Angle 1	Angle 2	Angle 3
4, 5, 7 cm			
8, 10, 14 cm			

SAS. Draw a 50° angle. Partner A makes its sides 3 and 5 cm; Partner B makes them 6 and 10 cm. Draw the third side and measure.

Triangle	Third side	Angle 2	Angle 3
50°, 3 and 5 cm			
50°, 6 and 10 cm			

What do you notice?

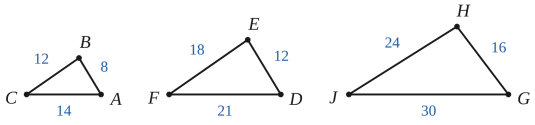
Part 2 · Notes and examples (fill in each note, then work the example)

SSS Similarity Theorem: if the corresponding _____ _____ of two triangles are proportional, then the triangles are similar.

Checking SSS: Compare shortest to shortest, _____ to _____, then the remaining pair. All the ratios must be _____.

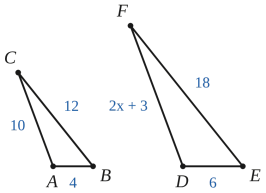
Example 1

Which of the triangles are similar? △ABC has sides 8, 12, 14. △DEF has sides 12, 18, 21. △GHJ has sides 16, 24, 30.



Example 2

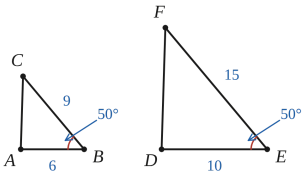
Find the value of x that makes △ABC ~ △DEF. AB = 4, BC = 12, AC = 10; DE = 6, EF = 18, DF = 2x + 3.



SAS Similarity Theorem: if an angle of one triangle is _____ to an angle of another and the lengths of the sides _____ these angles are proportional, then the triangles are similar.

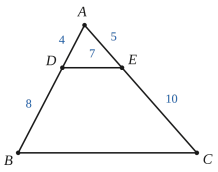
Example 3

△ABC has m∠B = 50°, AB = 6, and BC = 9. △DEF has m∠E = 50°, DE = 10, and EF = 15. Are the triangles similar? Explain.



Example 4

In △ABC, D is on $\overline{AB}$ and E is on $\overline{AC}$. AD = 4, DB = 8, AE = 5, and EC = 10. Show that △ADE ~ △ABC. Then find BC when DE = 7.



Example 5

△PQR has PQ = 6, QR = 8, and m∠R = 40°. △STU has ST = 9, TU = 12, and m∠U = 40°. Can you conclude that the triangles are similar?

Part 3 · Partner practice

Partner 1

△JKL has sides 6, 9, and 12. △MNP has sides 10, 15, and 20. Are the triangles similar? If so, give the scale factor from △JKL to △MNP.

Partner 2

In △ABC and △DEF, ∠A ≅ ∠D = 72°, AB = 8, AC = 12, DE = 6, and DF = 9. Are the triangles similar? Explain.